

DuPont™ Nomex® 410

Technical data sheet

Product description

DuPont™ Nomex® 410 is a family of insulation papers that offer high inherent dielectric strength, mechanical toughness, flexibility and resilience. Nomex® 410, the original form of Nomex® paper, is widely used in a majority of electrical equipment applications. Available in multiple thicknesses ranging from 0.05 mm to 0.76 mm (2 mil to 30 mil), Nomex® 410 is used in almost every known electrical sheet insulation application.

Electrical properties

The typical electrical property values for Nomex® 410 are shown in Table I. The AC rapid rise dielectric strength data in Table I represent voltage stress levels withstood for 10 to 20 seconds at a frequency of 60 Hz. These values differ from long-term strength potential. DuPont™ recommends that continuous stresses in dry type applications not exceed 1.6 kV/mm (40 V/mil) to minimize

the risk of partial discharges (corona). This recommendation is based on dielectric strength of the air and is only valid for dry type applications. The full wave impulse dielectric strength data shown in Table I are based on multiple sheets. These values are appropriate for applications that employ these materials in such configurations. The dielectric strength data are typical values and not recommended for design purposes. Design values can be supplied upon request.

Please note:

The properties in this data sheet are typical, or average values and should not be used as specification limits. Unless otherwise noted, all properties were measured in air under "standard" conditions (in equilibrium at 23°C, 50% relative humidity). Note that, like other products of papermaking technology, Nomex® papers have somewhat different properties in the papermaking machine direction (MD) compared to the cross direction (XD). In some applications it may be necessary to orient the paper in the optimum direction to obtain its maximum potential performance.

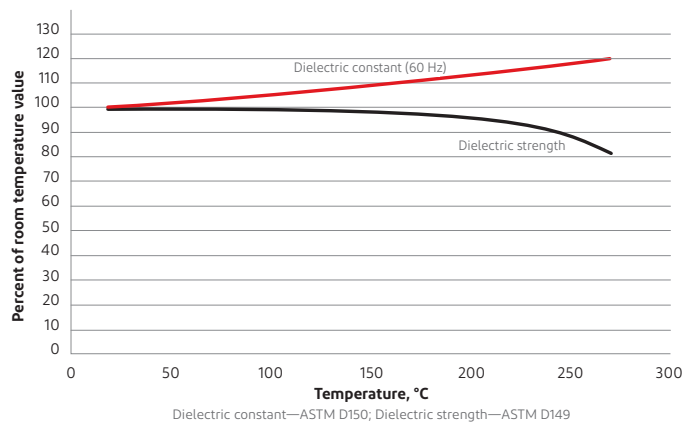
Table I. Typical electrical properties of Nomex® 410

Property	Nominal thickness, mm (mil)										Test method
	0.05 (2)	0.08 (3)	0.13 (5)	0.18 (7)	0.25 (10)	0.30 (12)	0.38 (15)	0.51 (20)	0.61 (24)	0.76 (30)	
Dielectric strength AC rapid rise,											
V/mil	460	565	715	865	845	870	850	810	810	680	ASTM D149 ¹
kV/mm	18	22	28	34	33	34	33	32	32	27	
Full wave impulse,											
V/mil	1000	1000	1400	1400	1600	N/A	1400	1400	N/A	1250	ASTM D3426
kV/mm	39	39	55	55	63	N/A	55	55	N/A	49	
Dielectric constant at 60 Hz	1.6	1.6	2.4	2.7	2.9	2.9	3.2	3.4	3.7	3.7	ASTM D150
Dissipation factor at 60 Hz (x 10 ⁻³)	4	5	6	6	6	7	7	7	7	7	ASTM D150

¹ Using 50-mm (2-in.) electrodes, rapid rise; corresponds with IEC 60243-1 subclause 9.1, except for electrode setup of 50 mm (2 in.).

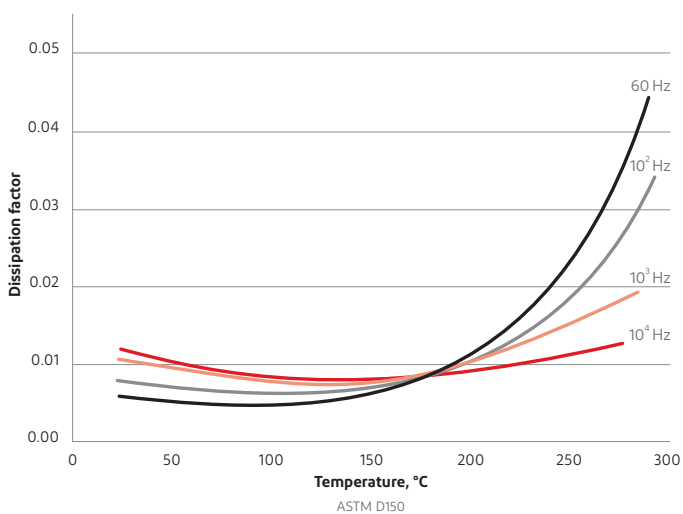
Temperature has a minor effect on dielectric strength and dielectric constant, as shown in Figure 1.

Figure 1. Effect of temperature on electrical properties of Nomex® 410—0.25 mm (10 mil)



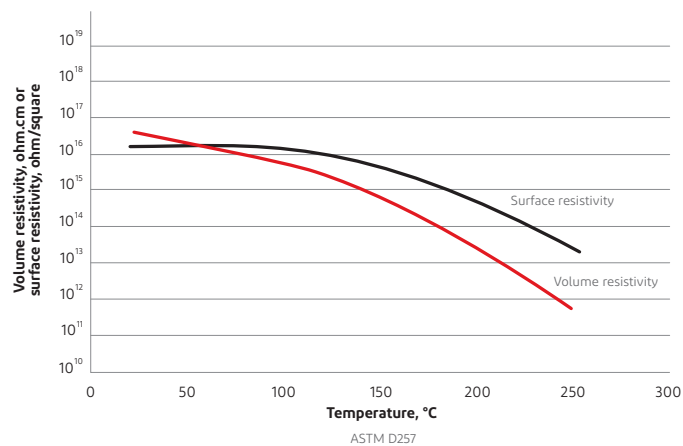
Variations in frequency up to 10 kHz have essentially no effect on the dielectric constant of Nomex® 410. The effects of temperature and frequency on the dissipation factor of dry Nomex® 410—0.25 mm (10 mil) paper are shown in Figure 2. The 60 Hz dissipation factors of thinner papers are essentially the same as those for 0.25 mm (10 mil) at temperatures up to 200°C. At higher temperatures and frequencies, the thicker papers have somewhat higher dissipation factors than those shown for 0.25 mm (10 mil) paper.

Figure 2. Dissipation factor versus temperature and frequency of Nome®x 410—0.25 mm (10 mil)



Surface and volume resistivities of dry Nomex® 410—0.25 mm (10 mil) paper are shown in Figure 3 as functions of temperature. The corresponding values for other thicknesses of Nomex® 410 are very similar.

Figure 3. Resistivity versus temperature of Nomex® 410—0.25 mm (10 mil)



The relatively minor effects of moisture (humidity) on the electrical properties of Nomex® 410—0.25 mm (10 mil) are shown in Table II.

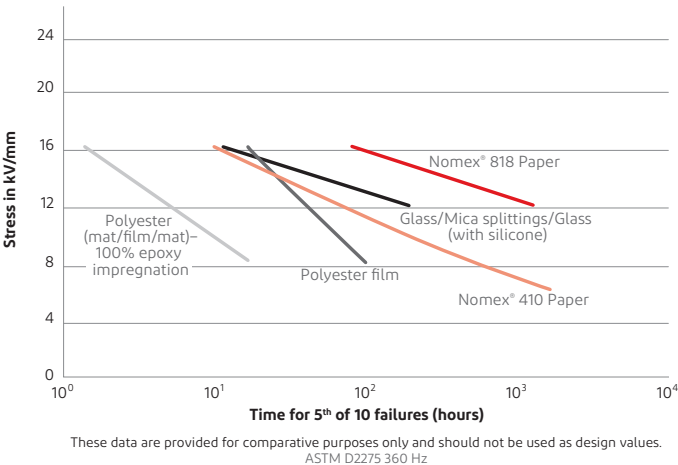
Table II. Humidity effects on electrical properties of Nomex® 410—0.25 mm (10 mil)

Property	Typical value			Test method
Relative humidity, %	Oven dry	50	96	
Dielectric strength				
V/mil	850	845	780	ASTM D149 ¹
kV/mm	33.5	33.2	30.7	
Dielectric constant				
at 60 Hz	2.5	2.7	3.2	ASTM D150
at 1 kHz	2.3	2.6	3.1	
Dissipation factor				
at 60 Hz (x 10 ⁻³)	6	6	11	ASTM D150
at 1 kHz (x 10 ⁻³)	13	14	25	
Volume resistivity				
ohm.cm	6 x 10 ¹⁶	2 x 10 ¹⁶	2 x 10 ¹⁴	ASTM D257

¹ Using 50-mm (2-in.) electrodes, rapid rise; corresponds with IEC 60243-1 subclause 9.1 except for electrode setup of 50 mm (2 in.).

Like other organic insulating materials, Nomex® paper is gradually eroded under attack by partial discharges. Partial discharge intensity is a function of voltage stress, which, in turn, depends almost entirely on design parameters such as spacing between circuit elements, smooth vs. sharp contours, etc. Although corona does not occur during normal operation of properly designed electrical equipment, any device may be subject to occasional over-voltages that produce brief corona discharges; and it is important that the insulation not fail prematurely under these conditions. The voltage endurance (time to failure under corona attack) of Nomex® 410 is superior to other commonly used organic insulations and even compares favorably with some inorganic compositions, as shown in Figure 4. These data were obtained in all cases on single layers of 0.25-mm (10-mil) materials at room temperature, 50% relative humidity and 360 Hz frequency. Times to failure at 50–60 Hz are approximately 6 to 7 times as long as indicated.

Figure 4. Voltage endurance of various insulating materials - single-layer Nomex® 410—0.25 mm (10 mil)



Mechanical properties

The typical mechanical property values for Nomex® 410 are shown in Table III

Table III. Typical mechanical properties of Nomex® 410

Property	Nominal thickness, mm (mil)										Test method
	0.05 (2)	0.08 (3)	0.13 (5)	0.18 (7)	0.25 (10)	0.30 (12)	0.38 (15)	0.51 (20)	0.61 (24)	0.76 (30)	
Typical thickness,											
mm	0.06	0.08	0.13	0.18	0.26	0.31	0.39	0.52	0.61	0.78	ASTM D374 ¹
mil	2.2	3.1	5.2	7.2	10.2	12.2	15.3	20.4	24.2	30.6	
Basis weight,											
g/m ²	41	64	115	174	249	310	395	549	692	839	ASTM D646
Density,											
g/cc	0.72	0.81	0.88	0.95	0.96	1.00	1.02	1.06	1.13	1.08	
Tensile strength, N/cm											
MD	43	68	141	227	296	380	462	610	728	816	ASTM D828
XD	19	34	71	116	161	185	252	374	500	592	
Elongation, %											
MD	10	12	16	20	22	23	22	23	21	21	ASTM D828
XD	7	9	13	15	18	18	16	18	16	17	
Elmendorf tear, N											
MD	0.8	1.2	2.3	3.7	5.6	7.1	9.0	14.3	N/A	N/A	TAPPI 414
XD	1.5	2.4	4.8	7.2	10.6	13.7	16.7	24.8	N/A	N/A	
Initial tear strength, N											
MD	11	16	31	48	69	88	110	158	191	233	ASTM D1004 ²
XD	6	9	17	27	42	55	71	114	153	193	
Shrinkage at 300°C, %											
MD	1.8	0.8	0.4	0.5	0.2	0.2	0.2	0.0	0.0	0.0	
XD	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.0	0.0	0.0	

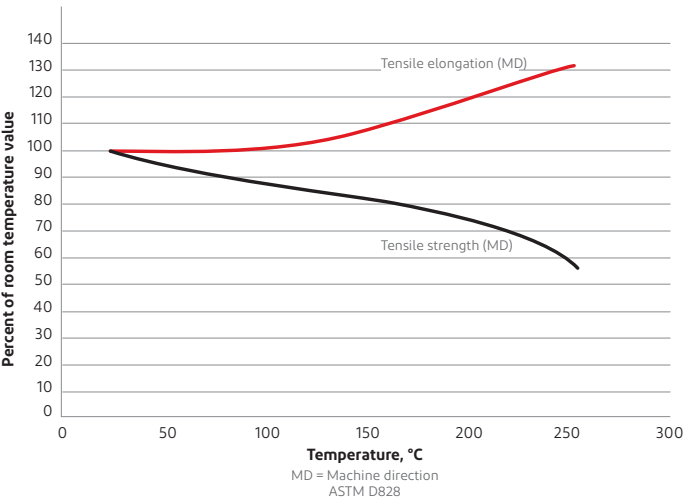
MD = machine direction; XD = cross direction

¹ Method D; using 17 N/cm².

² Data presented for initial tear strength are listed in the direction of the sample per ASTM D1004. The tear is 90 degrees to sample direction; hence, for papers with a higher reported MD initial tear strength, the paper will be tougher to tear in the XD.

The effects of high temperatures on tensile strength and elongation are illustrated in Figure 5. Nomex® sheet structures also retain good mechanical properties at very low temperatures. At the boiling point of liquid nitrogen (–196°C or 77K), the tensile strength of Nomex® 410—0.25 mm (10 mil) paper exceeds its room temperature value by 30% to 60% (depending on direction), while elongation to break is still greater than 3% (better than most inorganic materials at room temperature). This allows Nomex® 410 to work well in cryogenic applications.

Figure 5. Effect of temperature on mechanical properties of Nomex® 410—0.25 mm (10 mil)

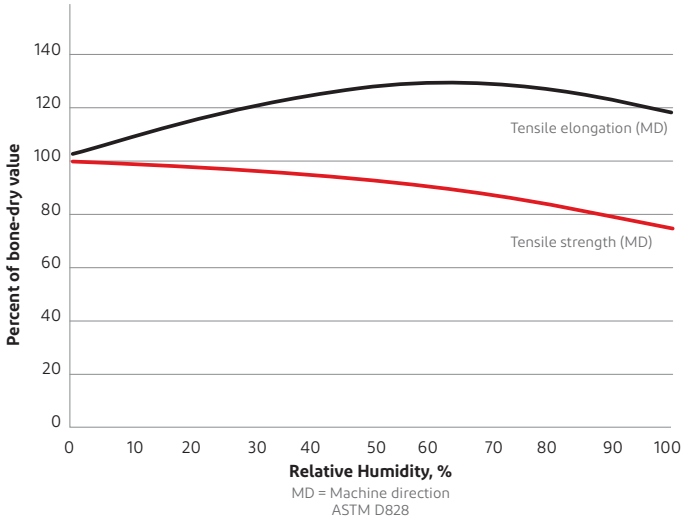


The effects of moisture (humidity) on tensile strength and elongation are shown in Figure 6. Like elongation, the tear strength and toughness of Nomex® 410 are also improved at higher moisture contents.

The dimensions of bone-dry Nomex® 410 exposed to 95% relative humidity (RH) conditions will increase at most by 1% in the machine direction and 2% in the cross direction (due to moisture absorption). This swelling is largely reversible when the paper is re-dried. The rate of change in dimensions will depend, of course, on paper thickness and configuration (e.g., individual sheets versus tightly wound rolls). Variations in environmental humidity will usually produce dimensional changes that will be less than 1%.

However, even small dimensional changes—especially if they are non-uniform—can cause or accentuate non-flatness (e.g., sag, puckers, etc.) in the sheet, which can cause problems in critical operations like laminating or creping. Therefore, Nomex® paper intended for these applications should be kept sealed in its protective polyethylene wrapper to maintain uniform moisture content until just before use. For more information on the effects of moisture on Nomex® insulation and how to appropriately protect the rolls, request our moisture brochure.

Figure 6. Effect of moisture on mechanical properties of Nomex® 410—0.25 mm (10 mil)



Thermal properties

The effects of long-time exposure of Nomex® 410—0.25 mm (10 mil) to high temperature on important electrical and mechanical properties are shown in Figures 7 and 8. These Arrhenius plots of aging behavior are the basis for the recognition of Nomex® paper as a 220°C insulation by Underwriters Laboratories (UL) and have been utilized for more than 50 years in commercial applications. These curves can also be extrapolated to higher temperatures. Measurements show, for example, that Nomex® 410 will maintain 300 V/mil (12 kV/mm) dielectric strength for several hours at 400°C, which is the performance predicted by the Arrhenius plot.

Figure 7. Useful life versus temperature for Nomex® 410—0.25 mm (10 mil) based on dielectric strength

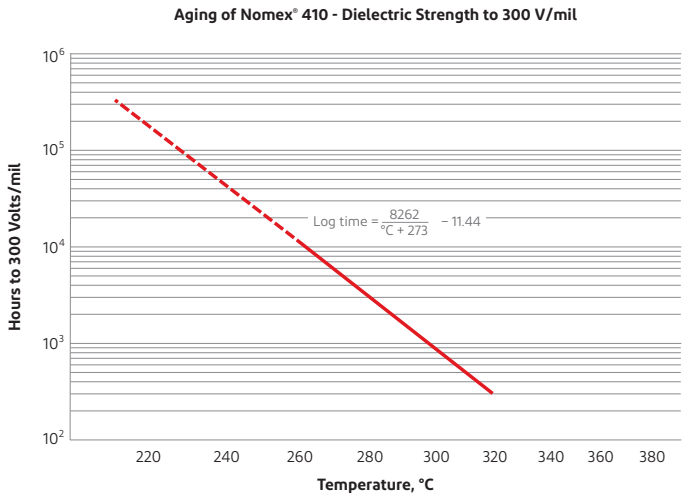
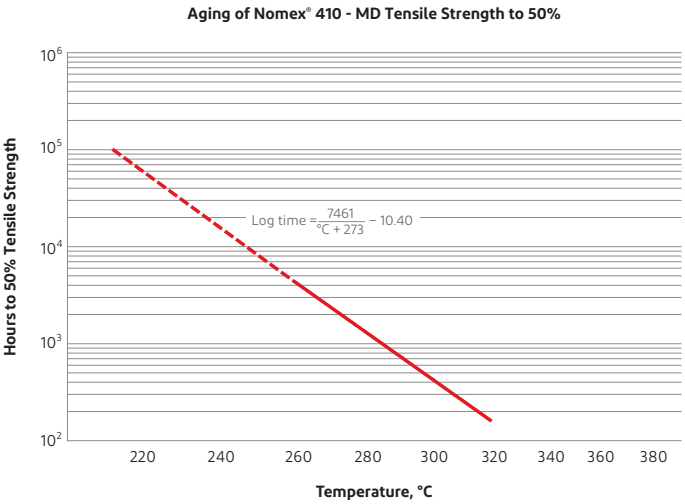


Figure 8. Useful life versus temperature for Nomex® 410—0.25 mm (10 mil) based on mechanical strength



The thermal conductivity of Nomex® 410—0.25 mm (10 mil) paper is shown in Figure 9. These values are similar to those for cellulosic papers and, as with most materials, are primarily determined by specific gravity (density). Therefore, thinner grades of Nomex® 410 will have slightly lower conductivity and thicker grades will have higher conductivities, as seen in Table IV. The total system construction may affect the overall thermal conductivity; therefore, care should be taken in applying individual sheet data to actual situations.

For example, two sheet insulations with identical thermal conductivities may have quite different effects on heat transfer from a coil due to the differences in stiffness or winding tension, which affect the spacing between the insulation layers.

Figure 9. Thermal conductivity versus temperature for Nomex® 410—0.25 mm (10 mil)

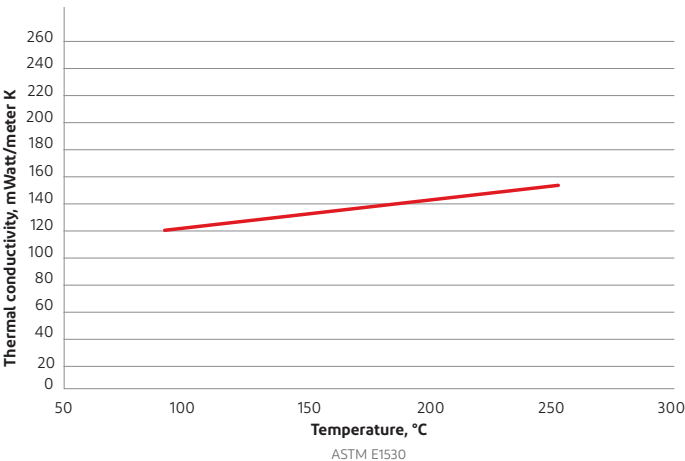


Table IV. Thermal conductivity of Nomex® 410

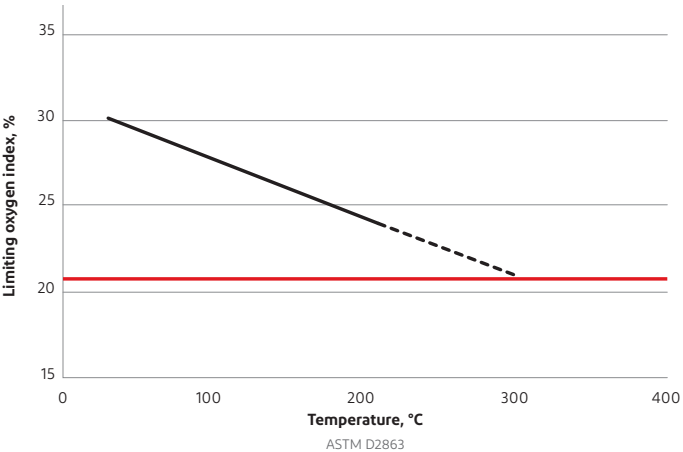
Property	Nominal thickness, mm (mil)							
	0.05 (2)	0.08 (3)	0.13 (5)	0.18 (7)	0.25 (10)	0.38 (15)	0.51 (20)	0.76 (30)
Density, g/cc	0.72	0.81	0.88	0.95	0.96	1.02	1.06	1.08
Thermal conductivity, ¹ mWatt/meter K	103	114	123	143	139	149	157	175

¹ All data taken at 150°C.
ASTM E1530

Chemical stability

The compatibility of Nomex® papers and pressboards with virtually all classes of electrical varnishes and adhesives (polyimides, silicones, epoxies, polyesters, acrylics, phenolics, synthetic rubbers, etc.), as well as with other components of electrical equipment, is demonstrated by the many UL recognized systems comprising Nomex®, including use in commercial applications for more than 50 years. “Nomex® papers are also fully compatible (and in commercial use) with transformer fluids (mineral and silicone oils as well as natural and synthetic esters) and with lubricating oils and refrigerants used in hermetic systems.” Nomex® papers also have recently been demonstrated as being the material of choice for insulating motors for electric vehicles, in large part due to demonstrated compatibility with fluids used in automotive applications such as automatic transmission fluids. Common industrial solvents (alcohols, ketones, acetone, toluene, xylene) have a slight softening and swelling effect on Nomex® 410, similar to that of water. These effects are mainly reversible when the solvent is removed.

Figure 10. Limiting oxygen index (LOI) for Nomex® 410—0.25 mm (10 mil)



Fire Performance

Nomex® 410 shows very good fire resistance performance. The performance has been confirmed by certification according to EN 45545-2 standard for fire protection on railway vehicles. Product suitability for the safety category Hazard Level 2 has been assigned based on measurements of LOI, smoke density and smoke toxicity.

Materials with LOI above 20.9% (ambient air) will not support combustion. Nomex® 410 must be heated between 240°C and 350°C (depending on thickness) before its LOI declines below the flammability threshold. The LOI data for Nomex® 410—0.25 mm (10 mil) are shown in Figure 10.

The fire performance has also been confirmed by 3rd party laboratory testing according to other recognized standards, like: ASTM E162, ASTM E1354, ASTM E662, BSS 7239. The required fire certification or RoHS information is available upon request.

Radiation resistance

The effect of 6400 megarads (64 Mgy) of 2 MeV beta radiation on the mechanical and electrical properties of Nomex® 410 is shown in Table V. (By comparison, a laminate of polyester film and polyester mat of the same thickness, 100% epoxy-impregnated, crumbled after 800 megarads, or 8 Mgy). Similar results were obtained on exposure to gamma radiation. The outstanding radiation resistance of Nomex® paper has led to its use in critical control equipment for nuclear power installations.

UL ratings

Table VI shows the UL ratings for Nomex® 410 papers. Descriptions of the numerical values for each of the UL ratings are detailed in the UL website on Component Materials, which can be accessed at in the UL iQ website under DuPont's UL file number E34739: iq.ul.com/ul/cert.aspx?ULID=230937.

Table V. Radiation resistance of Nomex® 410—0.25 mm (10 mil) to 2 MeV electrons (beta rays)

Property	Dose, Mgy								Test method
	0	1	2	4	8	16	32	64	
Tensile strength, % of original									
MD	100	96	100	100	94	87	81	65	ASTM D828
XD	100	100	99	99	97	86	81	69	
Elongation, % of original									
MD	100	89	92	96	76	60	36	18	ASTM D828
XD	100	92	91	88	82	47	27	16	
Dielectric strength, V/mil	860	860	840	840	840	860	890	790	ASTM D149 ¹
kV/mm	100	92	91	88	82	47	27	16	
Dielectric constant									
at 60 Hz	3.1	3.0	3.0	3.0	3.0	3.1	2.3	2.5	ASTM D150
at 1 kHz	3.0	3.0	2.9	3.0	2.9	3.1	2.3	2.5	
at 10 kHz	2.9	2.9	2.9	2.9	2.8	3.0	2.2	2.4	
Dissipation factor (x 10 ⁻³)									
at 60 Hz	8	14	10	12	9	14	7	10	ASTM D150
at 1 kHz	13	16	15	16	13	16	11	13	
at 10 kHz	18	21	20	20	19	20	15	17	

MD = machine direction; XD = cross direction
¹ With a 6.4-mm (1.4-in.) diameter electrode.

Table VI. UL ratings for Nomex® 410

Nominal thickness, mm	Nominal thickness, mil	UL94 flame class	UL746A HWI rating	UL746A HAI rating	UL746B RTI electrical	UL746B RTI mechanical	UL746A HVTR rating	UL746A CTI rating
0.05	2	—	0	3	220	220	3	3
0.08	3	—	0	3	220	220	3	3
0.13	5	V-0	0	1	220	220	3	3
0.18	7	V-0	0	1	220	220	3	3
0.25	10	V-0	0	0	220	220	3	3
0.30	12	V-0	0	0	220	220	3	3
0.38	15	V-0	0	0	220	220	3	3
0.51	20	V-0	0	0	220	220	3	3
0.61	24	V-0	0	0	220	220	3	3
0.76	30	V-0	0	0	220	220	3	3

Visit ei.nomex.com to learn more.

Product safety information is available upon request. This information corresponds to our current knowledge on the subject. It is offered solely to provide possible suggestions for your own experimentation. It is not intended, however, to substitute for any testing you may need to conduct to determine for yourself the suitability of our products for your particular purposes. This information may be subject to revision as new knowledge and experience become available. Since we cannot anticipate all variations in actual end-use conditions, DUPONT MAKES NO WARRANTIES AND ASSUMES NO LIABILITY IN CONNECTION WITH ANY USE OF THIS INFORMATION. Nothing in this publication is to be considered as a license to operate under or a recommendation to infringe any patent right. DuPont™, the DuPont Oval Logo, and all trademarks and service marks denoted with ™, SM or ® are owned by affiliates of DuPont de Nemours, Inc. unless otherwise noted. © 2024 DuPont. All rights reserved. (02/24).